

# CSA0313-DATA STRUCTURES

## CO-2\_Assessment-3

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### Linked List: Insert a Node at the Beginning (With Example)

#### Question 1: Create a new node

```
struct Node *newNode = (struct Node*)malloc(sizeof(struct Node));
```

**Example:** A new node is created in memory.

#### Question 2: Store the data in the new node

```
newNode->data = 30;
```

**Example:** The value **30** is stored in the new node.

#### Question 3: Make the new node's next point to the current head

```
newNode->next = head;
```

**Example:**

Current List: 10 -> 20 -> NULL

After linking: 30 -> 10 -> 20 -> NULL

#### Question 4: Update the head to the new node

```
head = newNode;
```

**Example:** Now the head points to node 30.

### Complete C Program

```
#include <stdio.h>
#include <stdlib.h>
```

```
struct Node{
    int data;
    struct Node *next;
};
```

```
struct Node *head=NULL;
```

```
void insertAtBeginning(int value){
    struct Node *newNode=(struct Node*)malloc(sizeof(struct Node));
    newNode->data=value;
    newNode->next=head;
    head=newNode;
}
```

```
void display(){
    struct Node *temp=head;
    while(temp!=NULL){
        printf("%d ",temp->data);
        temp=temp->next;
    }
}
```

```
int main(){
    insertAtBeginning(10);
    insertAtBeginning(20);
```

```
insertAtBeginning(30);  
printf("Linked List: ");  
display();  
return 0;  
}
```

**Program Output**

Linked List: 30 20 10